# Companion to Bird & de Moor, Algebra of Programming

# §4. Relations and Allegories

## §4.1. Allegories

## §4.2. Special properties of arrows

**Proposition 4.1**

`R:A→B, S:B→A, 𝟙 A⊑RS, SR⊑𝟙 B ⇒ S=R° and R is a map` A relation with a two-sided partial inverse under composition — identity below one order, identity above the other — is that inverse's converse, and R is itself a map.

## §4.3. Tabular allegories

**Proposition 4.2**

`(f,g) a tabulation of R ⇒ k°h⊑R ⇔ ∃! map m with h=mf, k=mg` Every pair below R's tabulating pair factors, uniquely, through a single map into the tabulation's apex.

## §4.4. Locally complete allegories

## §4.5. Boolean allegories

## §4.6. Power allegories